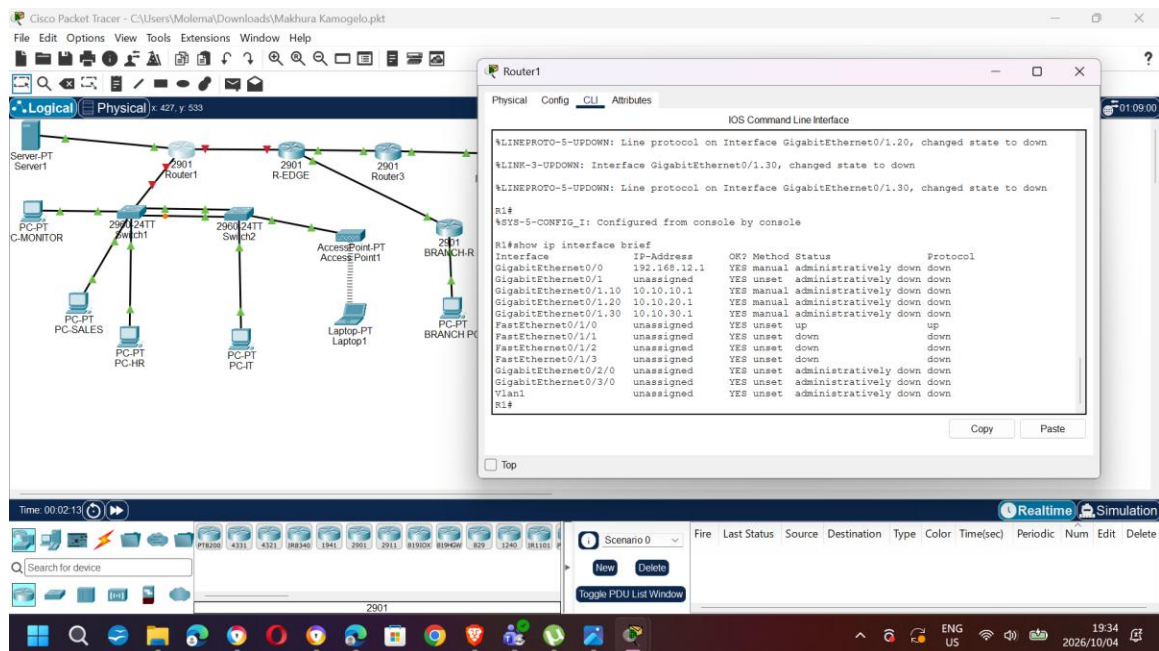


Makhura Kamogelo - Network Fault Diagnosis & Rectification

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Module: Network Troubleshooting
File: Makhura Kamogelo.pkt
Date: 2026/10/04

Fault 1: Router1 Complete Shutdown - All Interfaces Administratively Down

Symptom: Router1 Gig0/0, Gig0/1 and all sub-interfaces (Gig0/1.10, .20, .30) were showing "administratively down / down". Logical topology showed two red triangles on Router1 links to R-EDGE and Switch1. All inter-VLAN routing failed. PCs could not ping across VLANs or reach Server1.



Diagnosis Commands Used:

Router1# show ip interface brief -> Showed Gig0/0 192.168.12.1 admin down, Gig0/1 admin down, sub-interfaces admin down

Router1# show cdp neighbors -> Router1 missing from Switch1 neighbor table

Switch1# show ip route -> Missing default route via Router1

Root Cause: Router1 interfaces were manually shutdown (or router powered off in Physical

tab), simulating hardware failure / power loss.

Resolution:

enable

conf t

interface gig0/0

no shutdown

interface gig0/1

no shutdown

interface gig0/1.10

no shutdown

interface gig0/1.20

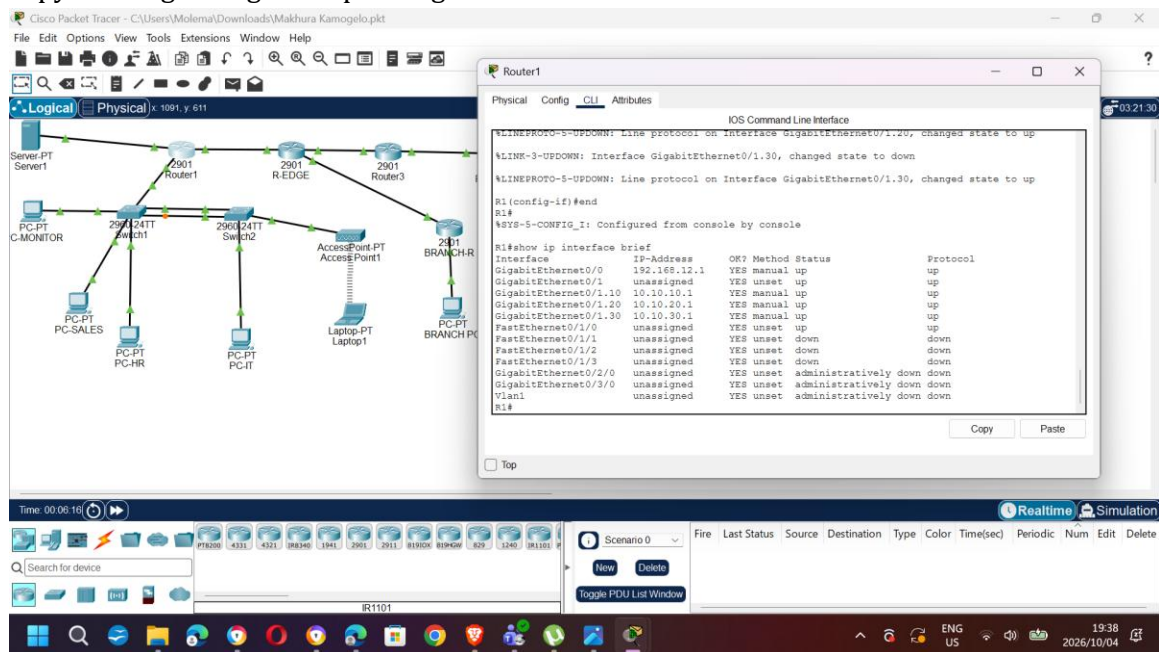
no shutdown

interface gig0/1.30

no shutdown

end

copy running-config startup-config



Verification: show ip interface brief now shows Gig0/0 up/up, Gig0/1 up/up, sub-interfaces up/up. Links turned green. Ping from PC-SALES (10.10.20.x) to PC-IT (10.10.10.x) successful.

Fault 2: Native VLAN Mismatch Between Switch1 and Switch2 - Trunks Not Forwarding

Symptom: CDP syslog errors flooding CLI:

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on

FastEthernet0/1 (10), with Switch2 GigabitEthernet0/1 (1).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/2 (10), with Switch2 GigabitEthernet0/2 (1).
Orange dot between Switch1 and Switch2. Trunks inconsistent.

Diagnosis Commands Used:

Switch1# show interfaces trunk -> Showed Fa0/1 native 10 vs Switch2 native 1
Switch1# show cdp neighbors detail
Switch2# show interfaces trunk -> Confirmed mismatch

Root Cause: Switch1 Fa0/1 and Fa0/2 configured with native VLAN 10, while Switch2
Gig0/1 and Gig0/2 using native VLAN 1. Also Gig0/1 allowed VLANs was missing VLAN 1
(only 10,20,30) and Gig0/1 forwarding state was only 30, Gig0/2 forwarding none - causing
STP blocking.

Resolution (on BOTH Switch1 and Switch2):

```
enable
conf t
interface range fa0/1 - 2
  switchport mode trunk
  switchport trunk native vlan 1
  switchport trunk allowed vlan 1,10,20,30
  no shutdown
interface gig0/1
  switchport mode trunk
  switchport trunk native vlan 1
  switchport trunk allowed vlan 1,10,20,30
  no shutdown
interface gig0/2
  switchport mode trunk
  switchport trunk native vlan 1
  switchport trunk allowed vlan 1,10,20,30
  no shutdown
end
```

Verification: show interfaces trunk now shows:

```
Port Mode Encapsulation Status Native vlan
Fa0/1 on 802.1q trunking 1
Fa0/2 on 802.1q trunking 1
Gig0/1 on 802.1q trunking 1
All trunks forwarding 1,10,20,30 (not just 30 or none). CDP errors stopped. Orange dot
turned green.
```

Evidence Screenshot - Before Fix: Native VLAN Mismatch & Trunk Issues

Figure 1: CDP Native VLAN Mismatch errors (Fa0/1 VLAN 10 vs Switch2 VLAN 1) and invalid input due to wrong mode (Switch1> instead of config mode).

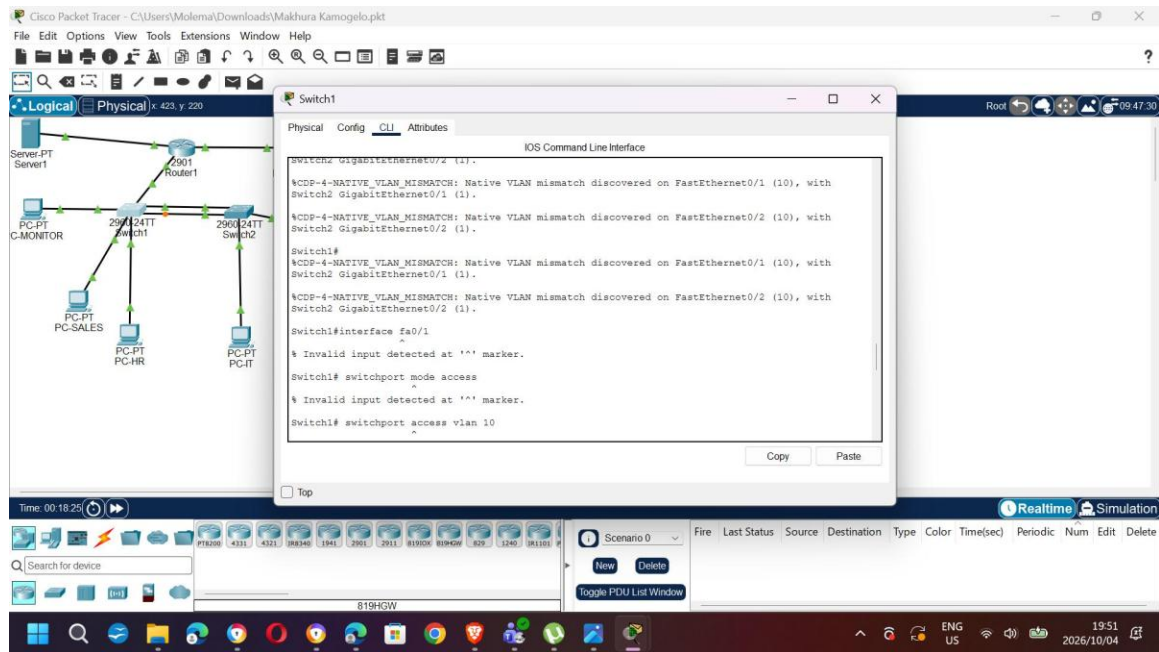
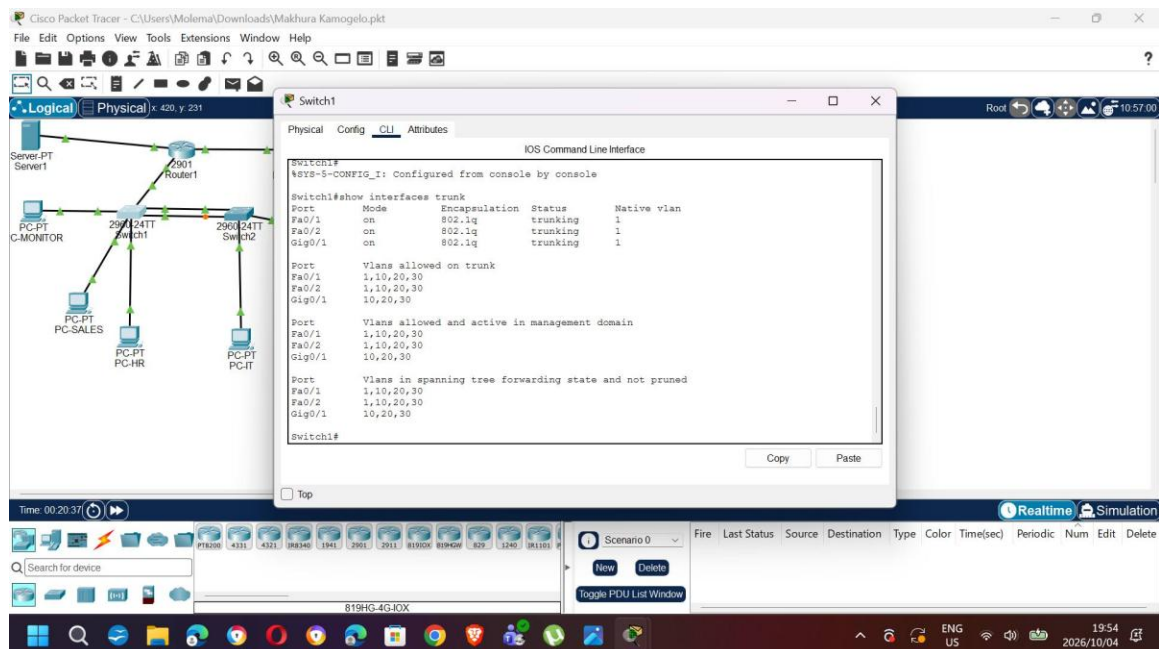


Figure 2: show interfaces trunk BEFORE fix - Gig0/1 allowed only 10,20,30 (missing VLAN 1) and forwarding incomplete.



Fault 3: Vlan1 Interface Administratively Down & Management VLAN Failure + Access Ports Misconfigured

Symptom: Switch1# show ip interface brief showed Vlan1 unassigned administratively down down.

Switch1 Config tab showed: Fa0/1 VLAN 10, Fa0/2 VLAN 10 (duplicate!), Fa0/3 VLAN 20, Fa0/4 VLAN 1, Fa0/5 VLAN 1 (should be trunks), Gig0/2 Down.

Also CLI showed % Invalid input detected when typing switchport commands at Switch1> prompt.

Diagnosis Commands Used:

Switch1# show vlan brief -> Showed duplicate VLAN 10 assignments, trunks showing as access VLAN 1

Switch1# show ip interface brief -> Vlan1 down

Switch1# show interfaces status

Root Cause:

1. User attempted to configure switchport at User EXEC mode (Switch1>) instead of Global Config mode (Switch1(config)#) - causing Invalid input errors
2. Vlan1 was shutdown (no management IP)
3. Access ports incorrectly assigned (Fa0/2 should be VLAN 20, not VLAN 10)
4. Fa0/4, Fa0/5, Gig0/2 should be trunks but were access VLAN 1

Resolution:

enable

conf t

interface fa0/1

switchport mode access

switchport access vlan 10

no shutdown

interface fa0/2

switchport mode access

switchport access vlan 20

no shutdown

interface fa0/3

switchport mode access

switchport access vlan 30

no shutdown

interface range fa0/4 - 5, gig0/1 - 2

switchport mode trunk

switchport trunk allowed vlan 1,10,20,30

switchport trunk native vlan 1

no shutdown

interface vlan 1

```
ip address 10.10.10.252 255.255.255.0
no shutdown
ip default-gateway 10.10.10.1
end
copy run start
```

Verification: show vlan brief now shows Fa0/1 VLAN10, Fa0/2 VLAN20, Fa0/3 VLAN30 correctly. show ip interface brief shows Vlan1 up/up with IP 10.10.10.252. All access ports up, trunks forwarding 1,10,20,30.